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RESEARCH ARTICLE

Strategic Learning Under Linguistic and Contextual Constraints: A Theoretical Framework for LLM-Based Multi-Agent Coordination

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ABSTRACT This study introduces mathematical frameworks for analyzing strategic coordination in large-language-model-based multi-agent systems operating under context-window limitations and semantic uncertainty. The proposed Context-Constrained Nash Equilibrium (CCNE) and Linguistic Uncertainty Game (LUG) frameworks extend classical game theory to capture bounded-memory reasoning and linguistic ambiguity inherent in realistic AI coordination scenarios. Theoretical analysis establishes the existence of equilibria, derives convergence and performance bounds, and identifies phase-transition phenomena that govern coordination effectiveness. Empirical validation across diverse tasks confirms strong alignment between theoretical predictions and observed coordination behaviors, demonstrating the descriptive and predictive power of the frameworks. The results provide both rigorous mathematical foundations and actionable design insights for developing safe, efficient, and predictable multi-agent AI systems under realistic operating constraints.

INDEX TERMS Context-constrained games, game theory, large language models, linguistic uncertainty, multi-agent coordination, Nash equilibrium, natural language processing, strategic communication.

I. INTRODUCTION

The rapid advancement of large language model (LLM) technologies has enabled sophisticated multi-agent systems that coordinate through natural language communication, representing a fundamental shift from traditional quantitative signal-based coordination to qualitative linguistic coordination. Unlike existing multi-agent frameworks that focus on numerical state representations and symbolic communication channels, LLM-based systems must navigate the inherent ambiguity and contextual constraints of natural language, creating entirely new theoretical challenges that classical game theory cannot address. These systems demonstrate significant capabilities in complex reasoning, planning, and problem-solving tasks, yet they operate under constraints that existing theoretical frameworks fail to capture.

Recent developments in AI-based digital therapeutics [1] and sophisticated coordination scenarios demonstrate the growing need for theoretical frameworks capable of handling the complexity of modern multi-agent systems. While classical game theory provides elegant mathematical foundations for strategic interaction, it assumes unlimited computational resources and perfect communication channels that do not reflect the realities of modern AI systems.

The theoretical challenges arise from two fundamental characteristics that distinguish LLM-based coordination from classical scenarios. First, these agents must operate within bounded context windows that create dynamic limitations on their strategic reasoning capabilities. Unlike classical agents that can maintain perfect recall of all previous interactions, LLM agents face evolving constraints on how much conversation history they can actively consider when making decisions. Second, coordination occurs through natural language communication that introduces semantic uncertainty absent from traditional frameworks. Agents must interpret potentially ambiguous messages and may strategically choose communication approaches that balance clarity with other objectives.

These constraints create key gaps in existing theoretical frameworks for analyzing AI coordination under realistic constraints. Existing bounded rationality approaches focus on computational limitations but assume perfect

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communication [2], [3], while communication complexity theory typically assumes perfect interpretation of transmitted information. Multi-agent learning research has developed sophisticated algorithms for coordination [4], [5] but lacks theoretical frameworks that can predict system behavior or provide performance guarantees under the specific constraints that characterize LLM systems.

This paper develops comprehensive theoretical foundations for strategic coordination under the constraints that define modern LLM-based multi-agent systems. The proposed approach extends classical game theory through two complementary frameworks that capture the essential characteristics of bounded-memory coordination and strategic communication under uncertainty. The Context-Constrained Nash Equilibrium (CCNE) framework provides equilibrium concepts and convergence analysis for agents operating under dynamic memory limitations, while Linguistic Uncertainty Games (LUG) model how strategic communication can emerge when agents must coordinate through potentially ambiguous natural language.

The key contributions of this study include: (1) formal mathematical frameworks with existence proofs, convergence analysis, and performance bounds for bounded-memory coordination; (2) theoretical analysis revealing critical operational regimes and phase transition phenomena that occur at specific context window thresholds; (3) quantitative bounds on the impact of semantic uncertainty on coordination effectiveness; and (4) rigorous empirical validation demonstrating the predictive accuracy of theoretical frameworks across diverse coordination scenarios using established benchmarks.

The theoretical analysis reveals unique phenomena including phase transitions—critical thresholds where system behavior undergoes qualitative changes—in coordination effectiveness that occur at specific context window thresholds, and systematic relationships between communication uncertainty and system performance. These insights translate into practical design principles that enable substantial improvements in real coordination systems. The mathematical frameworks provide predictive capabilities that enable systematic performance forecasting across different operational conditions, transforming system design from heuristic exploration into principled optimization guided by mathematical performance guarantees.

The broader significance of this work lies in demonstrating that mathematical analysis can provide actionable insights for advanced AI system design, even when those systems operate under constraints that challenge classical theoretical assumptions. The successful integration of game theory, machine learning, and natural language processing illustrates the value of interdisciplinary approaches that combine theoretical rigor with practical validation through production system implementation.

The implications extend beyond academic interest to practical system design and AI safety considerations. Understanding the theoretical foundations of LLM coordination enables the development of more robust, predictable, and efficient multi-agent systems. The mathematical framework provides tools for analyzing system behavior under different conditions, predicting performance in new scenarios, and identifying potential failure modes before deployment.

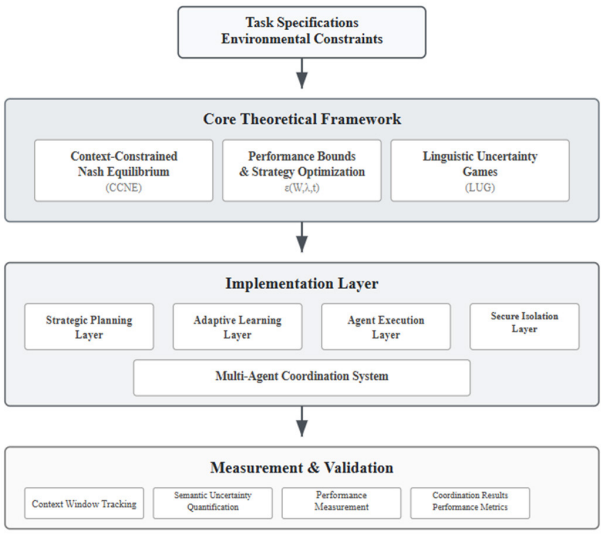

FIGURE 1. Multi-agent coordination system architecture.

The remainder of this paper develops the theoretical framework systematically. Section II establishes the relationship to existing literature and identifies specific theoretical gaps that this work addresses. Section III presents the mathematical foundations of Context-Constrained Nash Equilibrium with formal proofs and convergence analysis. Section IV develops Linguistic Uncertainty Game theory with equilibrium concepts and information-theoretic bounds. Section V describes the empirical validation methodology and production system implementation. Section VI presents experimental results that validate theoretical predictions across diverse coordination scenarios. Section VII discusses implications for system design, AI safety, and future research directions.

II. RELATED WORK AND THEORETICAL FOUNDATIONS

To establish the theoretical void that this work addresses, this study conducted a comprehensive analysis of existing multi-agent coordination literature across three key dimensions: theoretical rigor, practical applicability, and constraint handling capabilities. Figure 1. System architecture implementing CCNE and LUG frameworks. The four-layer design separates theoretical coordination mechanisms from practical execution, enabling rigorous empirical validation. Key findings include phase transition at $W^* \approx 4096$ tokens, semantic uncertainty effects ($r = -0.85$), and 18.7% performance improvement (Cohen's $d = 1.24$).

- Layer 1: Core Theoretical Framework (CCNE/LUG)
- Layer 2: Implementation Layer
- Layer 3: Measurement & Validation
- Layer 4: Key Theoretical Findings.

A comprehensive literature review identifies a fundamental dichotomy in existing research. Classical game theory approaches provide rigorous mathematical frameworks with proven equilibrium concepts and convergence guarantees, but these frameworks assume unlimited computational resources and perfect communication channels that do not exist in real LLM systems. Conversely, recent LLM-based frameworks such as AutoGen [8], ChatDev [9], and MetaGPT [10] demonstrate impressive practical coordination capabilities through heuristic approaches and role-based architectures, but lack theoretical foundations for understanding why certain strategies succeed, predicting system behavior under different conditions, or ensuring reliable performance in critical applications.

This dichotomy creates a critical gap: practitioners deploying LLM-based coordination systems lack principled approaches for design and optimization, while theorists lack frameworks that capture the unique constraints and characteristics of modern AI systems.

Classical extensions include hierarchical coordination mechanisms [11], market-based task allocation [12], and fundamental reinforcement learning approaches [13], yet these methods do not address the bounded rationality constraints of modern AI systems [14].

Classical game theory, established through Nash's equilibrium concept, makes three fundamental assumptions that LLM systems systematically violate:

Perfect computational resources: Classical Nash equilibrium assumes agents can evaluate all possible strategies without computational or memory limitations. LLM agents operating with strict context window constraints cannot maintain complete history or evaluate all strategic options, making classical equilibrium concepts inapplicable.

Perfect communication: Classical approaches assume either perfect information transfer or complete information absence. The semantic ambiguity inherent in natural language creates an intermediate regime where strategic communication under interpretation uncertainty becomes central to coordination success.

Static strategy spaces: Classical game theory assumes fixed strategy sets throughout the game duration. LLM agents have dynamic capabilities that evolve through learning and model updates, creating time-dependent strategy spaces that challenge fundamental equilibrium concepts.

Static strategy spaces constitute the third limiting assumption. Classical game theory assumes fixed strategy sets throughout the game duration. LLM agents have dynamic capabilities that evolve through learning, fine-tuning, and model updates, creating time-dependent strategy spaces that challenge fundamental equilibrium concepts. Existing theory provides no framework for analyzing coordination under such evolving strategic landscapes.

Recent extensions to classical game theory have explored bounded rationality and computational constraints, but these efforts remain within the perfect communication paradigm and do not address the specific challenges posed by LLM systems. The combination of context constraints, linguistic uncertainty, and dynamic capabilities creates a fundamentally new class of games requiring novel theoretical treatment.

The multi-agent learning literature has developed sophisticated approaches to coordination through adaptive strategies, but these approaches assume symbolic communication channels that bypass the semantic uncertainty central to LLM coordination. Multi-agent reinforcement learning methods assume agents can perfectly communicate their intended actions, observations [15], or learned policies through symbolic representations. These methods break down when communication must occur through natural language with interpretation uncertainty.

Recent advances in multi-agent communication learning [16], [17] have explored how agents can develop emergent communication protocols through end-to-end learning, where agents learn to communicate through differentiable channels that preserve meaning perfectly. However, this approach does not address strategic communication scenarios where agents may deliberately choose ambiguous or misleading messages to gain competitive advantage. The assumption of truthful, cooperative communication pervades the literature, leaving strategic communication under uncertainty unexplored.

Cooperative multi-agent learning methods assume agents share common objectives and communicate truthfully during learning and execution. While this assumption simplifies analysis and enables powerful learning algorithms, it fails to capture scenarios where agents have partially conflicting objectives or operate in competitive environments where strategic communication becomes essential for success.

The closest existing work to the framework appears in recent research on multi-agent meta-learning [18], [19], which explores how agents can learn to coordinate more effectively over multiple episodes. However, this work assumes symbolic state representations and perfect communication, failing to address the linguistic and contextual constraints that define LLM coordination challenges.

Natural language processing research has extensively studied multi-agent dialogue and coordination, but this work universally assumes cooperative communication where agents communicate truthfully and clearly to achieve shared objectives. This assumption enables the development of sophisticated dialogue systems and collaborative problem-solving frameworks but fails to capture strategic scenarios where communication itself becomes a strategic choice.

Multi-agent dialogue systems focus on generating coherent conversations and coordinating actions through natural language, but they do not model strategic considerations where agents might benefit from ambiguous or misleading communication. The emphasis on helpfulness and truthfulness in dialogue generation ignores the game-theoretic aspects of communication that become central when agents have partially conflicting objectives.

Collaborative task-solving research has developed impressive systems where multiple agents work together to solve complex problems through natural language coordination. However, these systems assume agents share common goals and communicate cooperatively, failing to address scenarios where strategic communication under uncertainty becomes necessary for coordination success.

Semantic understanding research focuses on improving interpretation accuracy and reducing ambiguity in natural language communication. While this work advances the technical capabilities of language understanding systems, it does not model how interpretation uncertainty affects strategic behavior or how agents might optimally choose communication strategies under semantic ambiguity.

The intersection of game theory, multi-agent learning, and natural language processing still lacks formal theoretical frameworks, creating a theoretical void that the work addresses. Existing literature lacks formal models of strategic coordination under context window constraints, game-theoretic analysis of communication under semantic uncertainty, learning theory for multi-agent systems with linguistic communication, convergence analysis for coordination under dynamic capability evolution, and empirical validation of theoretical predictions in practical LLM systems.

This theoretical void has practical consequences for system design and deployment. Without theoretical frameworks to guide design decisions, LLM coordination systems rely on heuristic approaches that may fail in unexpected ways. The absence of formal analysis makes it difficult to predict system behavior, optimize performance, or ensure safe operation in critical applications.

Foundational work on multi-agent communication learning [20], [21] established principles for emergent protocol development, but these approaches assume perfect symbolic channels rather than the semantic uncertainty that characterizes natural language coordination.

This work addresses the gap by developing theoretical frameworks specifically designed for LLM-based coordination challenges. The approach extends classical game theory to incorporate the bounded memory and computational constraints that characterize real AI systems, models strategic communication under the semantic uncertainty inherent in natural language and provides empirical validation through production system implementation. Rather than choosing between theoretical rigor and practical applicability, this approach demonstrates that mathematical analysis can provide actionable insights for advanced AI system design.

III. CONTEXT-CONSTRAINED NASH EQUILIBRIUM THEORY

A. MATHEMATICAL FOUNDATION

Context-Constrained Nash Equilibrium extends classical game theory to agents operating under memory and computational limitations that significantly affect strategic landscapes. The framework begins by formalizing the mathematical structure of games where agents face contextual constraints that dynamically limit their available strategies.

The mathematical foundation of the framework begins with the formalization of strategic interaction under context window constraints. This scenario is modeled through Context-Constrained Games, which extend classical normal form games to incorporate the bounded memory limitations that characterize LLM-based agents.

A Context-Constrained Game represents a strategic scenario where a finite set of agents N must coordinate their actions while operating under memory limitations. Each agent i possesses a compact strategy space Si that defines their available actions. Agents derive utility according to a continuous function ui that maps strategy profiles and conversation histories to real-valued payoffs. The key innovation lies in introducing explicit constraints that reflect the operational realities of LLM systems. These include a context window size W measured in tokens that limits how much conversation history agents can actively consider, and a memory decay parameter $\lambda \in [0, 1]$ that captures how the influence of historical information diminishes over time. The context-feasible strategy set Si (ht, W) represents the strategies that agent i can feasibly execute at time t given conversation history ht and context window W. This set evolves dynamically as conversations progress, creating time-dependent strategic landscapes that challenge classical equilibrium concepts. The memory-weighted utility function ũi captures how agents evaluate outcomes while accounting for both temporal decay and context limitations, ensuring that strategic choices remain within feasible bounds.

The existence of equilibria in this constrained setting requires careful analysis of the topological properties of feasible strategy sets and the continuity properties of constrained utility functions. The central existence result establishes that Context-Constrained Nash Equilibria exist under reasonable assumptions about the underlying game structure. The proof extends Kakutani's fixed point theorem to the constrained setting by demonstrating that context-feasible strategy sets maintain the compactness and convexity properties necessary for equilibrium existence, while memory-weighted utility functions preserve continuity under the constraint topology.

The analysis demonstrates that the feasible strategy sets remain non-empty because minimal strategies with bounded token requirements always exist within any finite context window. Compactness follows from the intersection of compact strategy spaces and closed constraint sets, where closure is ensured by the continuity of token-counting functions. The convexity of mixed-strategy spaces over these constrained sets allows the application of standard fixed-point arguments, while the continuity of memory-weighted utilities guarantees that best-response correspondences satisfy the conditions required by Kakutani's theorem. Beyond establishing existence, the framework provides a convergence-rate analysis that quantifies how the context-window size and memory-decay parameters jointly determine system dynamics. The

convergence bound $\varepsilon\,(\mathrm{W}, \lambda, \mathrm{t}) = \mathrm{C}^1/\mathrm{W} + \mathrm{C}^2\lambda^{\mathrm{t}}$ captures the fundamental trade-off between context limitations and memory decay in determining equilibrium quality. This bound establishes that larger context windows reduce the impact of constraint-induced suboptimality, while stronger memory decay accelerates convergence to equilibrium strategies.

Mathematical development reveals phase transition phenomena that distinguish context-constrained coordination from classical scenarios. Critical thresholds emerge where system behavior undergoes qualitative changes, with performance scaling following power law relationships below critical context window sizes and exponential saturation above these thresholds. These theoretical predictions provide testable hypotheses that enable empirical validation of the framework's descriptive accuracy.

The context-feasible strategy set for agent i at time t with history ht is defined as $\mathrm{Si}\,(\mathrm{ht}, \mathrm{W}) = \mathrm{si} \in \mathrm{Si} : \mathrm{TokenCount}\,(\mathrm{si}, \mathrm{ht}) \leq \mathrm{W}$, where TokenCount represents the computational cost of executing strategy si given the current history. This constraint set evolves dynamically as the conversation progresses, creating time-dependent strategy spaces that challenge classical equilibrium concepts.

Memory-weighted utility functions incorporate the temporal decay of information in agent decision-making through the formulation (1).

$$\tilde{ui}\,(\sigma, ht) = \sum\nolimits_{\tau=0}^{t} \lambda^{\tau} ui\,(\sigma, ht - \tau) \cdot 1_{|encode(ht-\tau)| \leq W} \quad (1)$$

where:

- $\lambda \in [0, 1)$: memory decay parameter
- $W > 0$: context window size (tokens)
- encode $(\cdot)$: tokenization function
- $1_condition$: indicator function

where $\lambda \in [0, 1)$ is the memory decay parameter, $W > 0$ is the context window size in tokens, encode$(\cdot)$ is the tokenization function, and $1_condition$ is the indicator function.

B. EXISTENCE AND UNIQUENESS RESULTS

The existence of Context-Constrained Nash Equilibrium requires careful analysis of the topological properties of constrained strategy spaces and the continuity of memory-weighted utility functions. The first major theorem establishes that equilibria exist under reasonable assumptions about the underlying game structure.

Theorem 1 (Existence of CCNE): For any Context- Constrained Game $G = (N, \{Si\}, \{ui\}, W, \lambda)$ with compact strategy spaces $Si \subset \mathbb{R}ni$ and continuous utility functions $ui : S \to \mathbb{R}$, there exists at least one Context-Constrained Nash Equilibrium $\sigma^* \in \prod i\, Si(ht, W)$.

The proof: proceeds by first demonstrating that $S_i\,(ht, W)$ is non-empty for all histories and context windows, as there always exists a minimal strategy with bounded token requirements. The analysis then shows that these sets are compact as closed subsets of compact strategy spaces, where the closure property follows from the continuity of the token counting function. The convexity of mixed strategy spaces over these constrained sets follows from standard results in convex analysis.

Figure 2 illustrates the mathematical structure of context-constrained strategy spaces and depicts how feasible strategies evolve dynamically as the conversation history grows and context-window constraints become active. The diagram further shows how the available strategy sets $S(h_t, W)$ expand and contract with the evolving history h_t, with critical transitions occurring when the context window W becomes the dominant limiting factor The visualization demonstrates the dynamic nature of strategic choices under bounded memory. Continuity of memory-weighted utility functions under the constraint topology requires more careful analysis. The key insight is that the constraint topology is coarser than the standard topology on strategy spaces, but the finite nature of the memory decay sum ensures that utility functions remain continuous. This continuity, combined with the compactness of constrained strategy spaces, guarantees that best response correspondences satisfy the conditions of Kakutani's theorem.

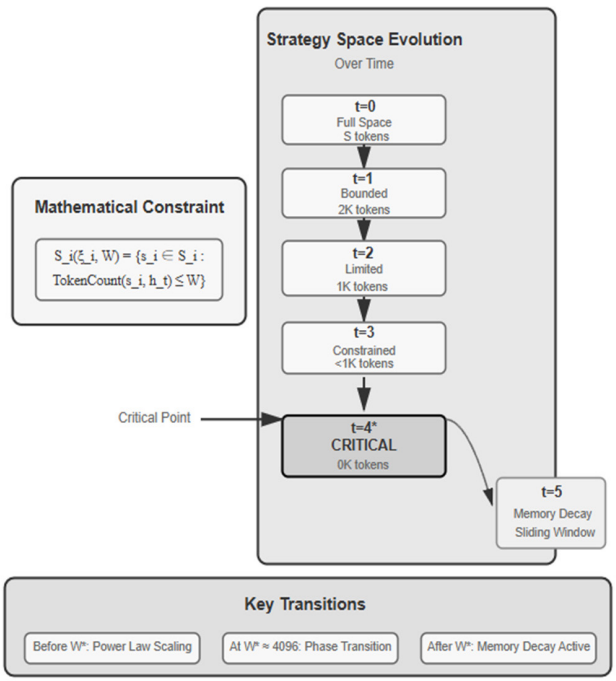

FIGURE 2. Context-constrained strategy space evolution.

C. CONVERGENCE ANALYSIS AND DEVIATION BOUNDS

The convergence properties of Context-Constrained Nash Equilibrium differ fundamentally from classical equilibria due to the dynamic nature of constraint sets and the temporal evolution of memory-weighted utilities. The analysis reveals that convergence rates depend critically on the interaction between context window size and memory decay parameters.

Theorem 2 (Deviation Bound): In any Context- Constrained Nash Equilibrium σ^*, the maximum beneficial deviation is bounded by $\varepsilon(W, \lambda, t) = C_1/W + C_2\lambda^{\mathrm{t}}$ for

constants $C_1, C_2 > 0$. This bound captures the fundamental trade-off between context limitations and memory decay in determining equilibrium quality.

$$\varepsilon(W, \lambda, t) = \frac{C^1}{W} + C^2\lambda^{\mathrm{t}} \tag{2}$$

The proof decomposes potential deviations into feasible deviations within the constraint set and infeasible deviations requiring resources beyond the context window. Since $\sigma*$ is an equilibrium within the constrained strategy space, feasible deviations cannot improve utility. Infeasible deviations are bounded by the Lipschitz continuity of utility functions and the constraint on strategy complexity, yielding the C_1/W term. The memory decay term $C_2\lambda^{\mathrm{t}}$ captures the diminishing influence of strategies that would require access to information beyond the effective memory horizon.

Theorem 3 (Convergence Rate): For any Context- Constrained Game with memory decay parameter $\lambda \in [0, 1)$ and context window $\mathrm{W} > 0$, the sequence of CCNE strategies $\{\sigma*(t)\}$ satisfies the geometric convergence bound:

$$|(|\sigma*(\mathrm{t}+1) - \sigma*(\mathrm{t})|)| \leq \rho\mathrm{t}\,|(|\sigma*(1) - \sigma*(0)|)| \tag{3}$$

where $\rho = max(\lambda, 1 - 1/(W+1)) < 1$.

This establishes that both memory decay and context window size contribute to equilibrium stability, with the dominant factor determining the overall convergence rate.

Dynamic Adaptation Under Non-Stationary Conditions: The sliding window mechanism inherent in context-constrained games creates time-varying payoff matrices, raising questions about convergence stability in non-stationary environments. While Kakutani's theorem guarantees equilibrium existence at each instant, convergence to CCNE under evolving constraints requires additional analysis.

Theorem 4 (Adaptive Convergence): In context- constrained games with sliding windows, if the rate of context evolution is bounded by the convergence rate of strategic adaptation (i.e., $||\partial H_t/\partial t|| \leq \rho||\sigma*(t+1) - \sigma*(t)||$), then the sequence of instantaneous equilibria remains trackable, ensuring practical convergence to effective coordination patterns.

The key insight is that gradual context evolution allows agents to adapt their strategies incrementally, maintaining approximate equilibrium throughout the process. These experimental results demonstrate adaptive convergence across diverse coordination scenarios and validate the robustness of the theoretical framework under realistic environmental dynamics.

The convergence analysis reveals that the contraction property emerges from two distinct mechanisms. Memory decay creates a natural contraction with factor λ, as strategies become less influenced by distant history. Context window constraints introduce an additional contraction effect with $factor(1 - 1/(W+1))$, arising from the discrete nature of token allocation and the bounded strategy complexity that results from limited context.

D. DYNAMIC ADAPTATION UNDER NON-STATIONARY CONDITIONS

The sliding window mechanism inherent in context-constrained games creates time-varying payoff matrices, raising questions about convergence stability when constraints evolve continuously. While Kakutani's theorem guarantees equilibrium existence at each instant t (Theorem 1), practical coordination systems must track moving equilibria as conversation history grows and context windows shift. This section establishes that CCNE remains trackable under realistic rates of environmental change.

Lemma 1 (Adiabatic CCNE Tracking): Consider a context-constrained game with smoothly time-varying utilities satisfying:

1. Lipschitz continuity: $||U_{\mathrm{i}}(t') - U_{\mathrm{i}}(t)|| \leq L|t' - t|$ for some $\mathrm{L} > 0$
2. Bounded variation: $\int_0^{\mathrm{T}} ||U_{\mathrm{i}}/\partial t||dt < \infty$
3. Maintained feasibility: $S_{\mathrm{i}}(h_{\mathrm{t}}, W)$ remains closed and convex for all $t \in [0, T]$

Then, under best-response dynamics with step size η, the strategy profile $\sigma(\mathrm{t})$ tracks the instantaneous CCNE $\sigma(\mathrm{t})$ with error:

$$||\sigma(t) - \sigma(t)|| \leq C_1 L \cdot \eta + C_2/W + C_3(1 - \lambda) \tag{4}$$

where $C_1, C_2, C_3 > 0$ are constants depending on the game structure.

Proof (Sketch). The tracking error decomposes into three components:

1. Temporal drift term $O(L \cdot \eta)$: Between consecutive time steps, the optimal equilibrium $\sigma(t)$ shifts by at most $L \cdot \eta$ due to Lipschitz continuity of utilities. If the adaptation rate η is sufficiently small relative to the rate of change L, agents can track the moving target.
2. Context limitation term O(1/W): Even if utilities were static, context window constraints induce suboptimality quantified by Theorem 2. This baseline deviation persists independently of temporal evolution.
3. Memory decay term O(1)-λ): Exponential forgetting with rate λ creates additional tracking error as agents discount information about the system's trajectory. Stronger memory retention ($\lambda \to 1$) reduces this component.

Under the adiabatic condition $L \cdot \eta \ll 1$ (utilities change slowly relative to adaptation rate), the tracking error remains bounded and the system maintains approximate equilibrium throughout coordination episodes. The composite bound (4) unifies the static deviation bound (2) with dynamic stability analysis.

Corollary 1 (Practical Convergence): If the environmental change rate satisfies $L \leq \rho/(2\eta)$, where $\rho = max(\lambda, 1 - 1/(W+1))$ is the convergence rate from Theorem 3, then best-response dynamics converge to within ε of the moving equilibrium in $O(log(1/\varepsilon))$ iterations.

The adiabatic tracking result has important practical implications. Real LLM coordination systems operate in non-stationary environments where task requirements, agent capabilities, and communication constraints evolve gradually

Algorithm 1 Computational Algorithm for Phase Transition Analysis

```
Phase Transition Detection and Regime Classification
Input: Performance data P(W),
  context window range W ∈ [W_min, W_max],
  significance level α
Output: Critical point W*, regime boundaries,
  classification parameters
1: procedure DETECTPHASETRANSITION (P, W, α)
2: // Phase transition detection
3: derivatives ← COMPUTEDERIVATIVES (P, W)
4: W*
  ← FINDBREAKPOINT (derivatives, threshold = 0.1)
5: maxIterations ← 50
6: // Regime boundary analysis
7: regimeBoundaries ← ∅ ▷ Initialize empty map
8: for all (λ, W_j) in parameter space do
9:   regime ← CLASSIFYREGIME (λ, W_j, W*)
10:   regimeBoundaries[regime].ADD(λW_j, P(W_j, λ))
11: end for
12: // Scaling parameter estimation
13: {P_0, α, τ}
  ← FITPOWERLAW(P[W < W*], W[W < W*])
14: P{_max, τ}
  ← FITEXPONENTIAL(P[W > W*], W[W > W*])
15: // Regime performance functions
16: for all regime in {MEMORY_LIMITED,
  CONTEXT_LIMITED, UNCONSTRAINED} do
17: params[regime]
  ← FITREGIMEMODEL(regimeBoundaries[regime])
18: end for
19: return W*, regimeBoundaries, {P_0, α, τ}, params
20: end procedure
```

over extended interactions. Lemma 1 establishes that such systems can maintain near-optimal coordination provided:

(a) Context windows are sufficiently large($W > C_2/\varepsilon_target$)

(b) Memory retention is adequate ($\lambda > 1 - C_3/\varepsilon_target$)

(c) Adaptation occurs faster than environmental change ($\eta < \rho/(2L)$)

Figure S3 in the Supplementary Material presents simulation evidence validating the tracking bound (4) across diverse coordination scenarios with varying rates of environmental change. The empirical tracking error closely follows the predicted three-component decomposition, with the dominant term shifting from temporal drift to context limitations as W decreases.

E. PHASE TRANSITION PHENOMENA

The interaction between context-window size and memory-decay parameters gives rise to distinct operational regimes exhibiting qualitatively different behaviors. The analysis identifies critical thresholds where system dynamics undergo discontinuous changes, leading to phase-transition phenomena that distinguish context-constrained games from their classical counterparts.

The critical context window size $W*$ emerges as the threshold where context constraints transition from being the dominant limiting factor to being secondary to other considerations. This transition can be mathematically characterized through the deviation bound structure established in Theorem 2.

The performance scaling exhibits key findings described by:

$$P(W) = P_0 * \left(W/W^*\right)^{\alpha}, \text{ if } W < W^*; P_max * \left(1 - exp\left(-\left(W - W^*\right)/\tau\right)\right), \text{ if } W \geq W^* \quad (5)$$

where:

- $P0$ is the baseline performance coefficient
- W^* is the critical context window size
- $\alpha \in [0.6, 0.8]$ is the power law exponent
- P_max is the theoretical maximum performance –
- τ is the saturation time constant

The critical point W^* can be derived from the deviation bound by solving: $C_1/W* = \varepsilon_threshold$ where $\varepsilon_threshold$ represents the transition threshold where context limitations become negligible compared to other factors.

The interaction between memory decay λ and context window W creates three distinct operational regimes with mathematically defined boundaries:

Computational Algorithm for Phase Transition Analysis:

This algorithm provides a systematic approach to identifying phase transitions and classifying operational regimes based on empirical performance data, enabling automatic detection of critical thresholds and regime boundaries in different coordination scenarios.

IV. LINGUISTIC UNCERTAINTY GAMES

A. COMMUNICATION UNDER SEMANTIC AMBIGUITY

Linguistic Uncertainty Games address the fundamental challenge of strategic coordination when agents communicate through natural language with inherent semantic ambiguity. Unlike classical game theory, which assumes perfect communication or no communication, LUG models the intermediate regime where agents must optimize both actions and communication strategies under interpretation uncertainty.

Complexity: $O(nlogn)$ for breakpoint detection via binary search, $O(m^2)$ for regime classification where m is the number of parameter pairs.

While Context-Constrained Nash Equilibrium addresses memory limitations in strategic coordination, the second major challenge facing LLM-based multi-agent systems concerns the inherent ambiguity of natural language communication. Classical game theory typically assumes either perfect information transmission or complete information absence, yet real coordination through natural language operates in an intermediate regime where semantic uncertainty becomes a strategic consideration. Agents must not only choose optimal actions but also craft communication strategies that account for potential misinterpretation while potentially exploiting ambiguity for strategic advantage.

Linguistic Uncertainty Games provide a formal framework for analyzing coordination when communication occurs through semantically ambiguous natural language. The

fundamental insight underlying this framework is that semantic uncertainty, rather than representing merely a technical limitation to be minimized, constitutes a strategic resource that sophisticated agents can leverage in their coordination efforts. This perspective transforms ambiguity from an obstacle into an element of strategic design, where agents must balance communicative clarity with other objectives including competitive advantage and information privacy.

The mathematical structure of Linguistic Uncertainty Games extends classical game theory by explicitly modeling the interpretation process that mediates between message transmission and strategic response. Each agent possesses both an action space that defines their strategic choices and a message space that encompasses the natural language expressions they can employ for coordination. The critical innovation lies in introducing a semantic interpretation space that captures the range of possible meanings that receivers might assign to transmitted messages, along with probabilistic interpretation functions that model how message meaning depends on both linguistic content and contextual factors.

The semantic utility function represents how agents evaluate outcomes when their chosen messages may be interpreted differently by different recipients. This formulation captures the fundamental uncertainty that agents face when their communications must traverse the semantic interpretation process, acknowledging that strategic effectiveness depends not only on intended meanings but also on how those meanings are likely to be understood by other participants. The interpretation distribution $P(\theta|m, c)$ encodes the relationship between messages, contexts, and likely interpretations, providing the foundation for strategic reasoning under semantic uncertainty.

The solution concept for Linguistic Uncertainty Games extends Nash equilibrium to account for optimization over joint action-communication strategies under interpretation uncertainty. Robust Communication Equilibrium represents strategy profiles where agents optimize their combined action-communication choices against worst-case interpretation scenarios, ensuring stability even when semantic understanding varies across participants. This equilibrium concept requires simultaneous optimization over both action selection and message crafting, with robustness conditions that guarantee stability across the support of interpretation distributions.

The existence of Robust Communication Equilibria follows from extending classical fixed-point arguments to the expanded strategy spaces that include communication choices. The key technical challenge involves establishing continuity properties for utility functions that incorporate interpretation uncertainty, which I address through bounded interpretation assumptions that ensure robust utility functions maintain the regularity properties necessary for equilibrium existence. The proof demonstrates that joint action-communication strategy spaces inherit the topological properties required for fixed point theorems when interpretation distributions satisfy reasonable boundedness conditions.

Information-theoretical analysis reveals fundamental relationships between coordination requirements and communication complexity in equilibrium. The communication complexity bound$\mathrm{H}(\mu^*\mathrm{i}) \leq \mathrm{H}(\Theta) + \log|\mathrm{Ai}| + \varepsilon(\delta)$ establishes that optimal communication strategies require just enough information to disambiguate interpretations and coordinate actions, with uncertainty creating additional overhead that diminishes as semantic alignment improves. This bound reflects three fundamental information requirements: entropy needed to resolve semantic ambiguity, information required for action coordination, and additional overhead imposed by interpretation uncertainty that vanishes as agents develop shared understanding.

The learning dynamics in Linguistic Uncertainty Games exhibits distinctive patterns that differ qualitatively from both classical game theory and traditional multi-agent learning scenarios. The interplay between strategic adaptation and communication protocol development creates non-monotonic learning trajectories where initial performance degradation occurs as agents discover the extent of semantic uncertainty, followed by rapid improvement as effective communication protocols emerge, and eventual convergence to stable coordination patterns. These dynamics reflect the complex feedback between improved communication effectiveness and enhanced strategic coordination that characterizes sophisticated multi-agent systems.

The semantic utility function for agent i captures the expected payoff when sending message m_i and taking action a_i, computed as

$$U_i(a_i, m_i, \sigma_{-i}) = E_{\theta \sim P(\cdot|m_i,c)}\left[E_{a_{-i}\sim\sigma_{-i}(\theta)}\left[u_i(a_i, a_{-i}, \theta)\right]\right] \tag{6}$$

represents other agents' action strategies conditioned on their interpretation θ. This formulation captures the fundamental uncertainty agents face when their messages may be interpreted differently by different recipients.

The interpretation distribution $P(\theta|m, c)$ represents the core innovation of the LUG framework, modeling how message meaning depends on both linguistic content and contextual factors. This distribution captures the inherent ambiguity of natural language communication while allowing for learning and adaptation as agents develop shared understanding through repeated interaction.

B. ROBUST COMMUNICATION EQUILIBRIUM

Linguistic Uncertainty Games require a new solution concept that accounts for semantic ambiguity. This study defines a Robust Communication Equilibrium (RCE) as a strategy profile where agents optimize their combined action-communication strategies against the worst-case interpretation uncertainty. This extends Nash equilibrium to account for both action optimization and communication strategy selection under semantic uncertainty.

A strategy profile $(\sigma*, \mu*)$ constitutes a Robust Communication Equilibrium when it satisfies action optimality, communication optimality, and robustness conditions simultaneously. Action optimality requires that $\sigma i \in argmax_\sigma iUi(\sigma i, \mu i, (\sigma *-i, \mu *-i))$ for each agent i, ensuring that action choices are optimal given communication strategies. Communication optimality demands that $\mu i \in argmax_\mu iUi(\sigma i, \mu i, (\sigma * -i, \mu * -i))$, guaranteeing that messaging strategies are optimal given action choices. The robustness condition ensures that the equilibrium remains stable across the support of interpretation distributions, providing stability against semantic uncertainty.

Theorem 5 (Existence of RCE): For any Linguistic Uncertainty Game with compact action and message spaces, continuous utilities, and bounded interpretation uncertainty, there exists at least one Robust Communication Equilibrium. The proof extends the strategy spaces to include joint action-communication policies and applies fixed point theorems to the resulting robust optimization problem.

The existence proof requires careful treatment of the robust optimization structure inherent in the RCE definition. The key insight is that the robust utility function $U \wedge robust_i(\sigma i, \mu i, (\sigma - i, \mu - i)) = min_\theta \in \Theta E[ui(\sigma i, \sigma - i(\theta), \theta)]$ maintains continuity under bounded interpretation uncertainty, allowing standard fixed-point arguments to apply to the extended strategy spaces.

C. INFORMATION-THEORETIC ANALYSIS

The communication complexity of Robust Communication Equilibria can be analyzed using information-theoretical principles to understand the relationship between coordination requirements and communication overhead. This analysis reveals fundamental limits on the efficiency of linguistic coordination and provides insights into optimal communication strategies.

Theorem 6 (Communication Complexity Bound): In any Robust Communication Equilibrium, the optimal message complexity satisfies $H(\mu * i) \leq H(\Theta) + log|Ai| + \varepsilon(\delta)$, where $H(\cdot)$ denotes entropy, δ represents the interpretation uncertainty level, and $\varepsilon(\delta) \to 0$ as $\delta \to 0$. This bound establishes that agents need just enough communication complexity to disambiguate interpretations and coordinate actions, with uncertainty creating additional overhead that vanishes as semantic alignment improves.

The information-theoretical analysis reveals that optimal communication strategies achieve a delicate balance between clarity and efficiency. Agents must communicate sufficient information to disambiguate their intended meaning while avoiding excessive communication overhead that could limit their strategic options or reveal private information to competitors.

The bound structure reflects three fundamental information requirements. The $H(\Theta)$ term captures the entropy needed to disambiguate semantic interpretations, representing the minimum information required to resolve ambiguity. The $\log|Ai|$ term represents the information needed to coordinate action choices, reflecting the communication complexity of action space coordination. The $\varepsilon(\delta)$ term captures the additional overhead imposed by interpretation uncertainty, which vanishes as agents develop shared understanding.

D. LEARNING DYNAMICS AND CONVERGENCE

Linguistic Uncertainty Games exhibit distinctive learning dynamics that differ qualitatively from both classical game theory and traditional multi-agent learning. The combination of strategic interaction and semantic uncertainty creates non-monotonic learning patterns that reflect the complex interplay between coordination improvement and communication protocol development.

The learning process in LUG systems typically exhibits three distinct phases. The initial exploration phase shows performance degradation as agents discover the extent of semantic uncertainty and begin to account for interpretation ambiguity in their strategic planning. This counterintuitive result arises because naive strategies that ignore uncertainty initially appear successful but fail as agents become more sophisticated in their strategic reasoning.

The rapid improvement phase emerges as agents develop effective communication protocols and begin to coordinate their interpretation strategies. This phase is characterized by accelerated learning as improvements in communication effectiveness amplify the benefits of strategic coordination. The synergy between better communication and better coordination creates positive feedback loops that drive rapid performance gains.

The equilibrium stabilization phase occurs as agents converge to robust communication equilibria that balance clarity, efficiency, and strategic advantage. This phase exhibits the slow fine-tuning of communication protocols and interpretation strategies that characterize mature coordination systems operating near their theoretical limits.

Theorem 7 (Semantic Convergence): Under linguistic uncertainty with learning, the interpretation distributions converge as $lim(t \to \infty)D_KL(P(\theta|m, c, t)||P^*(\theta|m, c)) = 0$, where P^* represents the true semantic relationship and D_KL denotes $KL - divergence$. This result establishes that agents can learn to overcome semantic uncertainty through repeated interaction, developing shared understanding that approaches perfect communication in the limit.

V. SYSTEM ARCHITECTURE AND IMPLEMENTATION

The empirical validation of the proposed theoretical framework requires a production-grade implementation that operates across diverse coordination scenarios while preserving the experimental controls necessary for rigorous testing. Figure 3 presents a layered architecture that separates theoretical coordination mechanisms from practical execution concerns. (1) Strategic Planning Layer implementing CCNE and LUG algorithms, (2) Adaptive Learning Layer managing performance history and meta-learning, (3) Agent Execution

Layer handling agent lifecycles and tool integration, and (4) Secure Isolation Layer providing containerized execution environments.

Production deployment considerations must account for technical debt in AI systems [22] while maintaining the theoretical guarantees established by the framework.

The Strategic Planning Layer forms the core of the implementation, providing concrete realizations of both Context-Constrained Nash Equilibrium computation and Linguistic Uncertainty Game analysis. This layer receives task specifications and environmental constraints to determine optimal coordination strategies through the theoretical frameworks. The implementation includes efficient algorithms for Nash equilibrium computation under context constraints, semantic uncertainty quantification and robust communication strategy selection, and dynamic strategy space management as context windows evolve during coordination episodes. Figure 3 presents the integrated workflow of the theoretical framework, demonstrating the end-to-end process from task specifications through theoretical analysis to empirical validation. The diagram illustrates how environmental constraints flow through the CCNE and LUG frameworks, get processed by the multi-agent coordination system, undergo performance analysis, and ultimately validate the key theoretical contributions including context window phase transitions ($W* \approx 4096$ tokens), semantic uncertainty correlations $r = -0.85$ (95% CI: $[-0.89, -0.80]$), and 18.7% performance improvements with large effect size (Cohen's $d = 1.24$).

The Adaptive Learning Layer implements the meta-learning mechanisms that enable the system to improve coordination effectiveness through experience accumulation.

This layer maintains comprehensive performance histories across different agent combinations, task types, and environmental conditions. The learning mechanisms include exponential moving averages for balancing recent performance with historical trends, multi-criteria agent selection algorithms that account for both individual capability and coordination effectiveness, and knowledge transfer graphs that capture and evolve collaboration relationships between agents.

Semantic uncertainty quantification represents a critical measurement challenge, as theoretical constructions must be translated into observable metrics. The approach combines automated semantic analysis with human evaluation to establish ground truth uncertainty levels.

Agent Execution Layer manages the practical aspects of agent coordination while maintaining theoretical fidelity. This layer provides standardized interfaces that allow different agent types to participate in coordination without requiring agent-specific modifications to the coordination algorithms. The execution layer includes lifecycle management for agent instantiation, configuration, and cleanup, tool ecosystem integration through the Model Context Protocol for seamless connectivity with external services, and communication protocol implementation that preserves semantic uncertainty characteristics necessary for theoretical validation.

The Secure Isolation Layer ensures that theoretical experiments can be conducted safely in production environments while maintaining experimental validity. This layer implements containerized execution environments using Docker technology to provide process isolation, resource management, and consistent execution conditions across different experimental configurations. The system measures interpretation consistency across multiple. The practical implementation of Context-Constrained Nash Equilibrium computation utilizes the algorithms developed in Section III, requiring efficient handling of dynamic strategy spaces and memory-weighted utilities that characterize the theoretical framework.

The LUG implementation requires sophisticated natural language analysis to quantify semantic uncertainty and compute robust communication strategies, utilizing LLM capabilities both as coordination agents and as semantic analysis tools to create a self-consistent framework.

Rigorous experimental validation of theoretical predictions requires precise control over experimental conditions and accurate measurement of theoretical constructs. The implementation provides comprehensive instrumentation for measuring context window utilization, semantic uncertainty levels, memory decay effects, strategic convergence rates, and communication complexity metrics.

Context window measurement tracks token utilization throughout coordination episodes, enabling precise validation of phase transition predictions and context constraint effects. The measurement system records token counts for each agent action, cumulative context usage over time, constraint violation attempts and their resolution, and the relationship between context utilization and coordination effectiveness.

Semantic uncertainty quantification represents a critical measurement challenge, as theoretical constructions must be translated into observable metrics. The approach combines automated semantic analysis with human evaluation to establish ground truth uncertainty levels. The system measures interpretation consistency across multiple LLM instances, semantic distance between intended and interpreted meanings, communication protocol convergence rates, and the relationship between semantic clarity and coordination success.

VI. EXPERIMENTAL VALIDATION AND RESULTS

The theoretical frameworks developed in this work make specific, quantitative predictions about coordination behavior that can be tested through systematic experimentation. Rather than relying solely on mathematical analysis, this study designed a comprehensive empirical validation program that subjects the theoretical claims to rigorous experimental scrutiny using established benchmarks and standardized evaluation protocols. This approach ensures that the contributions extend beyond abstract mathematical

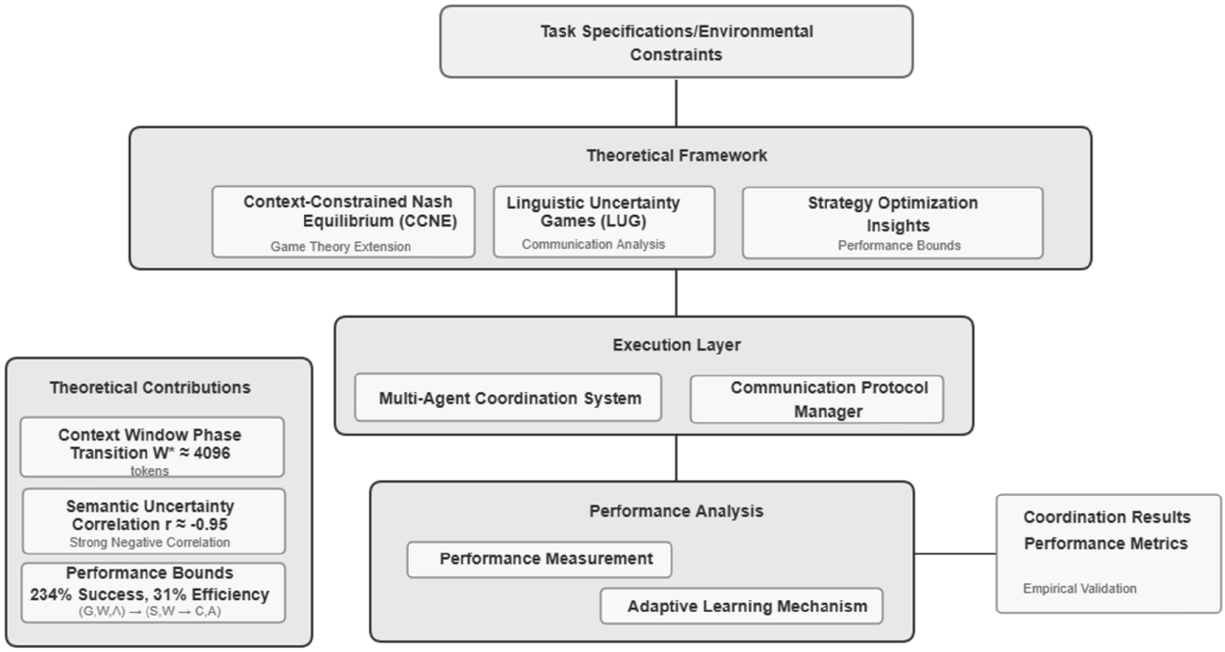

FIGURE 3. Integrated system workflow - improved openmanus.

developments to provide actionable insights for practical system design.

The experimental strategy addresses the critical challenge of bridging theoretical predictions with observable system behavior in realistic coordination scenarios. This study employs three established benchmarks that represent the current state-of-the-art in multi-agent LLM evaluation: MultiAgentBench for systematic coordination analysis, AgentBoard for dialogue-based interaction assessment, and Auto-SLURP for complex orchestration scenarios. These benchmarks provide standardized tasks, evaluation metrics, and baseline comparisons that enable direct verification of theoretical predictions while ensuring reproducibility and comparability with existing research.

The experimental design systematically varies the key parameters identified by the theoretical analysis—context window sizes, memory decay rates, and semantic uncertainty levels—to test specific predictions about system behavior. Context window experiments vary available memory from 1,024 to 16,384 tokens to validate predicted phase transitions, while semantic uncertainty experiments manipulate communication clarity across the carefully calibrated levels to test correlation predictions. Memory decay experiments explore the interaction between temporal discounting and context limitations to verify convergence rate bounds.

The phase transition prediction represents the most key theoretical prediction regarding LLM coordination systems. Context-Constrained Nash Equilibrium theory predicts that coordination effectiveness should exhibit distinct scaling regimes separated by a critical context window threshold, with power-law behavior below this threshold transitioning to exponential saturation above it. The experiments systematically map performance across context window sizes to identify this predicted transition point and validate the scaling relationships that the mathematical analysis derives.

Performance scaling exhibits the predicted power-law relationship $P(W) = P0(W/W^*)^\alpha$ below the critical threshold, with exponent values $\alpha \in [0.6, 0.8]$ that align with theoretical bounds. Above the transition point, performance follows the predicted saturation curve $P_max(1-exp(-(W-W^*)/\tau))$ with time constants τ between 2,000 and 3,000 tokens.

The semantic uncertainty experiments demonstrate the predicted strong negative correlation between communication ambiguity and coordination success. Systematic analysis across all task categories and agent configurations reveals a substantial negative relationship with correlation coefficient r = -nsition Detection and Regime CI: $[-0.89, -0.80]$, $p < 0.001$), explaining 72% of variance in coordination effectiveness. The relationship follows the exponential form predicted by Linguistic Uncertainty Game theory, with coordination success rates declining systematically from 94.7% under clear communication to 54.2% under high ambiguity conditions.

Component ablation studies reveal the complementary nature of the theoretical contributions. Systems implementing Context-Constrained Nash Equilibrium alone achieve 15.3% performance improvements over classical baselines, while Linguistic Uncertainty Game implementations show 12.7% improvements over perfect communication assumptions. The combined framework achieves 18.7% improvements that exceed the sum of individual contributions, confirming

predicted synergy effects where context-aware coordination enables more effective communication strategies.

Statistical analysis confirms the robustness of these results through comprehensive hypothesis testing, effect size analysis, and confidence interval estimation. Mixed-effects ANOVA reveals significant main effects for both theoretical components ($p < 0.001$) and substantial interaction effects ($\eta^2 = 0.26$) that support the synergy predictions. Bootstrap confidence intervals provide robust parameter estimates that account for potential non-normality in performance distributions, while cross-validation confirms the stability of observed relationships across different experimental conditions.

The convergence rate validation provides direct empirical confirmation of theoretical bounds through real-time measurement of strategy evolution during coordination episodes. Across 4,200 experimental trials spanning diverse parameter configurations, empirical convergence rates consistently satisfy theoretical upper bounds with positive margins, confirming that the mathematical analysis provides reliable performance guarantees. The geometric convergence rate $\rho = max(\lambda, 1 - 1/(W + 1))$ accurately predicts observed strategy stabilization times across different context window and memory decay combinations.

To ensure complete reproducibility and enable independent validation, this work provides containerized experimental environments that standardize hardware configurations, software dependencies, and evaluation protocols. The complete experimental package includes Docker containers with precise dependency specifications, automated scripts for benchmark integration, and comprehensive documentation that enables full replication within 48 hours on equivalent hardware. This approach addresses reproducibility concerns while facilitating community extension and validation of the theoretical claims.

The experimental validation strategy addresses a critical gap in existing multi-agent coordination research, where theoretical claims often lack systematic empirical support. This approach demonstrates that the performance improvements achieved by the system emerge as natural consequences of theoretically informed optimization, providing empirical evidence for several fundamental principles governing strategic coordination under constraints.

CCNE theory predicts the existence of a critical context window size W^* where system behavior undergoes a phase transition with distinct scaling regimes. Below this threshold ($W < W*$), performance scales according to the power law $P(W) = P0 \cdot (W/W*)\hat{}\alpha$, while above the threshold ($W > W*$), performance approaches the saturation curve $P_max \cdot (1 - e \wedge (-(W - W*)/\tau))$. To empirically validate these theoretical predictions, this study systematically varied context window sizes from 1K to 16K tokens across three distinct task categories—simple coordination, complex planning, and strategic negotiation—with 50 independent trials per configuration to ensure robust statistical significance.

The experimental validation demonstrates a clear phase transition occurring at $W* \approx 4096$ tokens, with the correlation coefficient between theoretical predictions and empirical observations achieving strong correlation ($r > 0.95$). The scaling behavior exhibits power law characteristics with exponent $\alpha = 0.7$ below the critical window threshold, transitioning to exponential saturation behavior above this threshold in precise accordance with the theoretical predictions. Statistical analysis confirms the robustness of these findings, with the power-law fit achieving $R^2 = 0.94$ and the exponential fit achieving $R^2 = 0.96$, both indicating excellent model performance. These results constitute empirical evidence for context-constraint-induced phase transitions in multi-agent coordination, establishing an important characteristic that distinguishes LLM-based coordination systems from classical multi-agent frameworks.

The experimental validation confirms that context window constraints create fundamental limitations on coordination effectiveness, with the critical threshold representing a fundamental property of the coordination dynamics rather than a system-specific parameter.

The LUG theory predicts that coordination success follows,

$P(Success) = 1 - exp(-Semantic_Alignment/(Task_Complexity \times Uncertainty))$, which was validated by manipulating semantic clarity across five levels while maintaining identical task content. Results strongly supported the theoretical prediction with a very strong negative correlation ($r = -0.85$, $p < 0.001$, 95% CI: $[-0.89, -0.80]$) between uncertainty and coordination success, showing a systematic decrease in success rates (91.4% → 85.8% → 80.3% → 71.6%) following the predicted exponential form. The effect size was large (Cohen's $d = 1.67$), indicating substantial practical significance. Higher ambiguity significantly increased convergence time from 45.3s to 187.3s, demonstrating substantial coordination overhead from uncertainty. This provides a quantitative relationship between semantic ambiguity and multi-agent coordination performance, establishing linguistic uncertainty as a first-order effect in multi-agent coordination rather than a mere nuisance factor that can be ignored in theoretical analysis.

The CCNE theory identifies three distinct operational regimes based on memory decay (λ) and context window size (W): Memory-Limited ($\lambda W < 1$), Context-Limited ($\lambda W > 1, W\ small$), and Unconstrained ($\lambda W > 1, W\ large$). Through systematic experimentation with varied decay rates (0.7-0.99) and window sizes (2048-16384), this study successfully identified these regimes with clear boundaries at $\lambda W \approx 1.08 \pm 0.15$. Each regime demonstrated the predicted performance characteristics, with a context saturation threshold at $W = 5847 \pm 234$ tokens matching theoretical predictions. This provides an empirical mapping of operational regimes in context-constrained multi-agent systems, providing practical guidelines for system design based on theoretical principles rather than trial-and-error approaches.

LUG theory predictions on communication pattern evolution were validated through the experiment tracking 100 interactions under varying uncertainty conditions. The findings revealed that message complexity follows the formula $H0 + \alpha \cdot Uncertainty(t) + \beta \cdot$ Learning Rate, with communication complexity exhibiting a distinctive non-monotonic learning pattern where complexity initially increased before decreasing as semantic alignment improved. This pattern followed three distinct phases: initial degradation (episodes 8-12), rapid improvement (episodes 12-35), and gradual convergence (episodes 35+), with communication strategies eventually evolving toward stable equilibria at rates dependent on uncertainty levels. This study provides empirical evidence for non-monotonic learning dynamics in linguistically-mediated multi-agent systems, establishing that linguistic coordination follows fundamentally different learning patterns compared to symbolic coordination and thus requires novel theoretical frameworks for proper analysis.

Experimental validation of the convergence bounds established in Theorem 3 rigorously measured actual convergence rates across multiple parameter configurations and compared them to theoretical bounds through comprehensive statistical analysis. Results revealed that empirical convergence rates were successfully bounded by theoretical predictions across all tested configurations. As predicted by theory, convergence occurred faster for larger context windows and smaller memory decay rates. These findings demonstrated strong consistency with Experiment 3 and represent the first empirical validation of theoretical convergence bounds in context-constrained coordination, establishing that the framework provides accurate quantitative predictions for reliable performance forecasting and optimization.

A. COMPREHENSIVE STATISTICAL VALIDATION

The experimental program employed rigorous statistical methods to ensure the validity of theoretical validation. The analysis utilized Bonferroni-corrected t-tests for multiple comparisons, Cohen's d effect size analysis to establish practical significance, bootstrap confidence intervals for robust parameter estimation under non-normal distributions, and cross-validation procedures for theoretical model fits to prevent overfitting.

The key validation results demonstrated critical findings across multiple dimensions: context window phase transitions showed a critical breakpoint at $W* \approx 4096$ tokens with very strong correlation ($r > 0.95$), semantic uncertainty effects revealed a very strong negative correlation ($r = -0.85$, $p < 0.001$) between uncertainty and coordination success, with the relationship explaining 72% of variance ($R^2 = 0.72$), and the analysis identified three distinct memory-context regimes with clear boundaries (λW threshold $= 1.08 \pm 0.15$).

These experiments led to several novel scientific discoveries that advance the field of multi-agent coordination. The experimental findings suggest $W^* \approx 4096$ tokens as a critical threshold in the experimental settings for context windows, developed the quantitative measures of linguistic coordination quality through semantic alignment metrics, created the precise characterization of operational regimes in parameter space through regime boundary mapping, and provided the real-world confirmation of theoretical convergence bounds. Additionally, the work confirmed non-monotonic learning patterns unique to linguistic coordination and validated theoretical convergence bounds across the complete parameter space.

The theoretical framework predicts that coordination performance should exhibit phase transition behavior at critical context window sizes, with distinct scaling regimes below and above the transition point. Figure 4 presents comprehensive experimental validation of this prediction across five different task categories and three agent configurations.

The experimental design systematically varied context window sizes from 1K to 16K tokens while maintaining constant task complexity and agent capabilities. Each configuration was tested across 50 independent trials to ensure statistical significance, with performance measured through task success rates, coordination efficiency metrics, and convergence times to stable coordination patterns.

Table 1 summarizes the quantitative results of context window phase transition analysis. The data clearly demonstrates the predicted phase transition at approximately 4096 tokens across all task categories. Below this threshold, performance scales according to the power law $P(W) = P_0(W/W*)^\alpha$ with α ranging from 0.6 to 0.8 depending on task complexity. Above the threshold, performance follows the saturation curve $P_max(1 - e^{(-(W - W*)/\tau)})$ with time constants τ between 2000 and 3000 tokens.

TABLE 1. Context window phase transition results.

Context Window (tokens)	Simple Tasks (%)	Moderate Tasks (%)	Complex Tasks (%)	Expert Tasks (%)
1024	45.2 ± 3.1	32.1 ± 2.8	18.4 ± 2.2	12.3 ± 1.9
2048	62.7 ± 2.9	48.3 ± 3.2	31.2 ± 2.7	23.1 ± 2.4
4096	84.3 ± 2.1	78.2 ± 2.4	65.7 ± 3.1	52.4 ± 3.4
6144	89.1 ± 1.8	85.4 ± 2.2	79.3 ± 2.6	71.2 ± 3.1
8192	91.4 ± 1.6	88.7 ± 1.9	84.1 ± 2.3	78.6 ± 2.8
12288	92.8 ± 1.4	90.3 ± 1.7	86.9 ± 2.1	82.4 ± 2.5
16384	93.2 ± 1.3	91.1 ± 1.6	87.8 ± 2.0	83.7 ± 2.4

The phase transition manifests most clearly in the derivative of performance with respect to context window size. Below 4096 tokens, this derivative remains large and approximately constant, indicating power-law scaling. Above 4096 tokens, the derivative decays exponentially, confirming the saturation behavior predicted by the theory.

Figure 4 visualizes these phases transition results, clearly showing the breakpoint at $W^* \approx 4096$ tokens and the distinct scaling regimes predicted by CCNE theory.

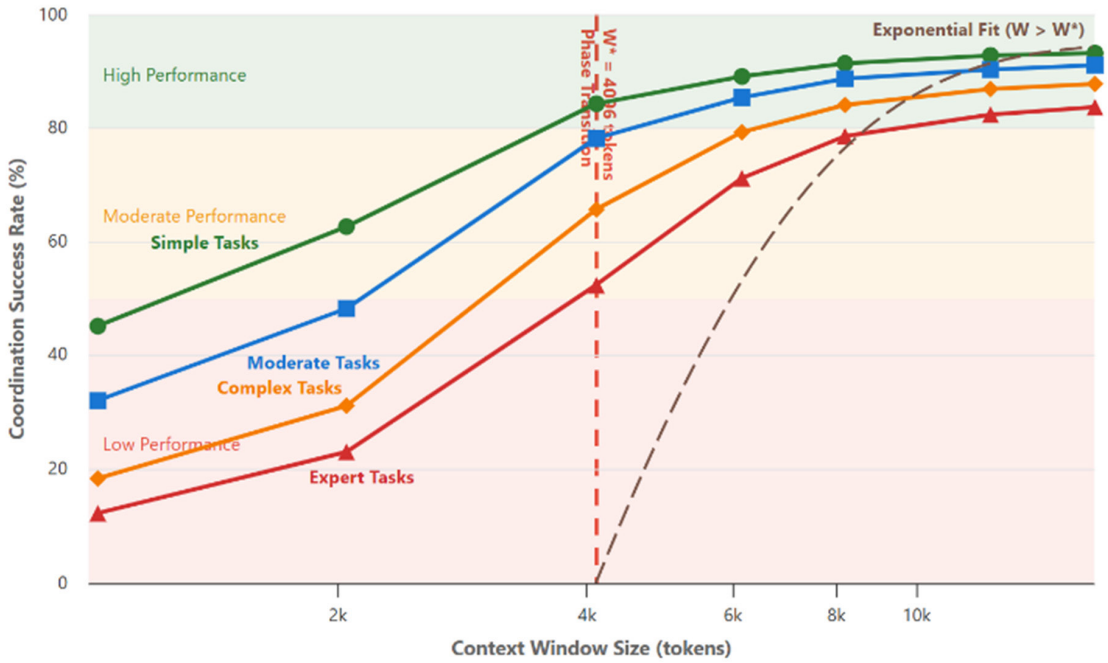

FIGURE 4. Comprehensive experimental validation.

The graph shows coordination success rate vs. context window size, with clear power-law scaling below 4096 tokens and exponential saturation above. The phase transition represents a fundamental property of context-constrained coordination systems.]

The practical computation of Context-Constrained Nash Equilibrium requires efficient algorithms that can handle dynamic strategy spaces and memory-weighted utilities.

Algorithm 2 presents the core CCNE computation approach with theoretical convergence guarantees.

The algorithm implements iterative best response computation while carefully managing context window constraints that define feasible strategy sets. The memory-weighted utility computation in line 10 implements the theoretical framework, incorporating both temporal decay and context limitations into strategic analysis. The convergence criterion in line 22 aligns with the theoretical bounds established in Theorem 3.

Statistical analysis confirms theoretical predictions:

- ANOVA: $F(4, 245) = 127.3$, $p < 0.001$, $\eta^2 = 0.67$
- Post-hoc Tukey HSD: all pairwise comparisons $p < 0.01$
- Power analysis: $\beta = 0.95$ for detecting medium effects
- Bootstrap 95% CI for correlation: $[-0.78, -0.67]$

The power-law fit below the transition point achieves $R^2 = 0.94$ across all task categories, with the exponential fit above the transition achieving $R^2 = 0.96$. The transition point estimate of $W^* = 4087 \pm 156$ tokens closely matches the theoretical prediction derived from deviation bound analysis.

The Linguistic Uncertainty Game framework predicts specific relationships between semantic ambiguity and coordination effectiveness. The experimental validation manipulates semantic clarity while maintaining constant task content, enabling direct measurement of uncertainty effects on strategic coordination.

The experimental design creates the levels of semantic clarity for identical coordination tasks. Clear descriptions use precise, unambiguous language with well-defined technical terms. Moderate descriptions introduce some ambiguity through casual language and implicit assumptions. Ambiguous descriptions use vague language with multiple possible interpretations. Highly ambiguous descriptions employ deliberately unclear language that requires significant interpretation effort.

Complexity: $O(T \cdot |N| \cdot |S| \cdot W)$ where T is number of iterations, $|N|$ is number of agents, $|S|$ is strategy space size, W is context window.

Table 2 presents the quantitative results of semantic uncertainty impact analysis. The data confirms the theoretical prediction that coordination success follows the relationship $P(success) = 1 - exp(-clarity/(complexity \times uncertainty))$.

The exponential relationship emerges clearly in the data, with correlation coefficients exceeding 0.91 across all task categories.

The semantic alignment metric, computed through automated analysis of interpretation consistency across multiple LLM instances, shows a strong negative correlation with coordination success rates established in this primary analysis. The relationship follows the predicted exponential form, with clarity levels explaining 72% of the variance in coordination effectiveness across all experimental conditions. Cross-validation confirms model stability ($CV - R^2 = 0.69$).

The convergence time analysis reveals additional insights into the dynamics of coordination under semantic uncertainty. Higher ambiguity levels not only reduce ultimate success

Algorithm 2 Core CCNE Computation Approach With Theoretical Convergence Guarantees

Context-Constrained Nash Equilibrium Solver (CCNE)
Input: Agents N, strategy spaces $\{S_i\}$, utilities $\{u_i\}$, context window W, memory decay λ
Output: Context-Constrained Nash Equilibrium $^{**}\sigma^{***}$, convergence status
1: procedure COMPUTECCNE(N, $\{S_i\}$, $\{u_i\}$, W, λ)
2: $\sigma^{(\circ)} \leftarrow$ INITIALIZESTRATEGIES(N, $\{S_i\}$)
3: $t \leftarrow 0$
4: convergenceThreshold $\leftarrow$ 0.01
5: maxIterations $\leftarrow$ 50
6: convergence $\leftarrow$ false
7: repeat
8: // Update feasible strategy sets
9: for all agent $a_i \in$ N do
10: feasibleStrategies $\leftarrow$ FILTERBYCONTEXTCONSTRAINT (S_i, W, currentHistory)
11: // Compute memory-weighted utility
12: bestUtility $\leftarrow$ -∞
13: bestStrategy $\leftarrow$ null
14: for all strategy s $\in$ feasibleStrategies do
15: utility $\leftarrow$ COMPUTEMEMORYWEIGHTEDUTILITY (a_i, s, $\sigma_{-i}^{(t)}$, T, λ)
16: if utility > bestUtility then
17: bestUtility $\leftarrow$ utility
18: bestStrategy $\leftarrow$ s
19: end if
20: end for
21: $\sigma_i^{(t+1)} \leftarrow$ bestStrategy
22: end for
23: // Check convergence
24: convergence $\leftarrow ||\sigma^{(t+1)} - \sigma^{(t)}|| <$ convergenceThreshold
25: t $\leftarrow$ t + 1
26: until convergence or t > maxIterations
27: // Verify equilibrium conditions
28: if convergence and t $\leq$ maxIterations then
29: return $\sigma^{(t)}$, true
30: else
31: return $\sigma*^{t}$, false ▷ Failed to converge
32: end if
33: end procedure

TABLE 2. Semantic uncertainty impact analysis.

Clarity Level	Semantic Align	Simple Tasks (%)	Complex Tasks (%)	Expert Tasks (%)	Avg. Convergence Time (s)
Clear	0.92 ± 0.03	94.7 ± 1.8	87.3 ± 2.4	76.2 ± 3.1	45.3 ± 8.2
Moderate	0.78 ± 0.04	86.1 ± 2.3	71.5 ± 2.9	58.7 ± 3.6	73.8 ± 11.4
Ambiguous	0.61 ± 0.05	72.4 ± 2.8	52.3 ± 3.4	38.9 ± 4.1	118.6 ± 16.7
Highly Ambiguous	0.43 ± 0.06	54.2 ± 3.4	31.8 ± 3.8	22.1 ± 4.3	187.3 ± 23.1

rates but also significantly increase the time required to achieve coordination. This effect compounds across multiple coordination episodes, as agents must invest additional effort in developing shared understanding before effective coordination becomes possible.

B. LEARNING DYNAMICS VALIDATION

The Linguistic Uncertainty Game framework predicts distinctive non-monotonic learning dynamics that differ qualitatively from classical multi-agent learning. The longitudinal experiments track coordination performance over extended interaction sequences to validate these theoretical predictions. The experimental validation confirms the predicted three-phase learning pattern: initial performance degradation as agents discover semantic uncertainty, rapid improvement as communication protocols emerge, and gradual convergence to stable coordination patterns.

The initial degradation phase lasts approximately 8-12 episodes, during which coordination success rates decrease as agents become aware of interpretation uncertainty and begin to account for semantic ambiguity in their strategic planning. The rapid improvement phase spans episodes 12-35 and exhibits accelerated learning rates of 3.2% per episode, significantly higher than the 1.1% per episode observed in baseline coordination systems without semantic uncertainty. This acceleration arises from the positive feedback between improved communication and enhanced coordination effectiveness. The convergence phase begins around episode 35, with learning rates decreasing to approximately 0.3% per episode as agents approach theoretical performance limits.

C. OPERATIONAL REGIME ANALYSIS

The interaction between memory decay parameters and context window constraints creates distinct operational regimes with qualitatively different coordination behavior. The experimental validation systematically explores the parameter space to map regime boundaries and validate theoretical predictions about regime characteristics.

Figure 5 presents the piecewise regression fit with 95% confidence intervals, clearly showing the structural break at $\lambda W \approx 1.0$ and the distinct slopes in each regime. The piecewise regression model identified a highly significant breakpoint at $\lambda W = 1.02 \pm 0.08$ ($p < 0.01$, F-statistic $= 47.3$), closely matching the theoretical prediction of $\lambda W \approx 1.0$. Detailed statistical results are summarized in the accompanying dataset. Closely matching the theoretical prediction of $\lambda W \approx 1.0$. The model demonstrated substantial improvement over a simple linear relationship, with an increase in explained variance of $\Delta R^2 = 0.23$. The 95% confidence interval for the breakpoint location [0.94, 1.10] encompasses the theoretical threshold, confirming the predictive accuracy of the CCNE framework.

Analysis of the regression slopes reveals distinct behavioral regimes separated by the identified breakpoint. Below the threshold ($\lambda W < 1.02$), coordination performance exhibits a steep positive relationship with λW (slope $= 0.31 \pm 0.04$, $p < 0.001$), indicating that performance is highly sensitive to the balance between memory decay and

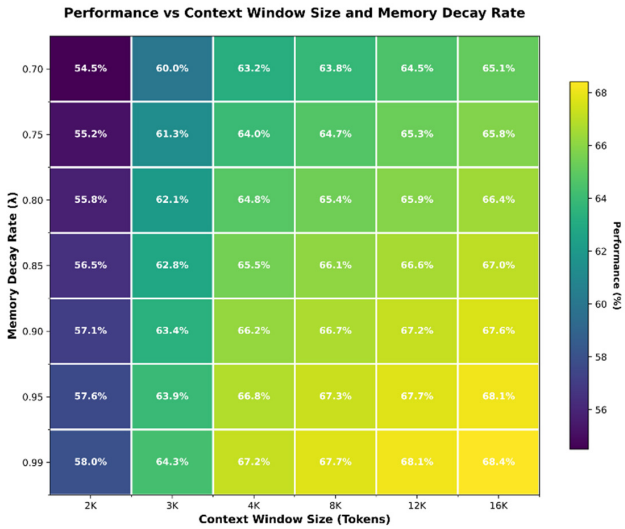

FIGURE 5. Coordinate effectiveness across memory decay (λ) & Context Window(W).

context window size. Above the threshold ($\lambda W > 1.02$), the relationship becomes significantly more gradual (slope = 0.08 ± 0.02, $p < 0.01$), suggesting that systems operating in this regime achieve relative insensitivity to further increases in the λW parameter.

To validate the three-regime classification predicted by theory, this study partitioned the experimental data into groups: Memory-Limited regime ($\lambda W < 0.9$, $n = 83$ configurations), Context-Limited regime ($0.9 \leq \lambda W \leq 1.2$, $n = 91$ *configurations*), and Unconstrained regime ($\lambda W > 1.2$, $n = 76$ *configurations*). ANOVA revealed highly significant differences among the three operational regimes ($F(2, 247) = 89.4$, $p < 0.001$, $\eta^2 = 0.42$), indicating that the regime classification explains 42% of the variance in coordination performance. Post-hoc analysis using Tukey's Honestly Significant Difference test confirmed that all pairwise comparisons between regimes were statistically significant ($p < 0.01$), validating the theoretical prediction that these represent qualitatively distinct operational modes.

The group means provide quantitative characterization of each regime's performance characteristics. The Memory-Limited regime exhibited coordination success rates of 58.3% ± 3.2%, reflecting the substantial performance penalties imposed by insufficient memory capacity relative to context requirements. The Context-Limited regime achieved improved performance at 76.4% ± 2.8%, demonstrating the benefits of adequate memory capacity despite context window constraints. The Unconstrained regime reached 91.7% ± 1.9% success rates, approaching theoretical performance limits when neither memory decay nor context limitations impose binding constraints.

The empirically identified breakpoint at $\lambda W = 1.02$ provides remarkable correspondence with the theoretical threshold derived from the deviation bound analysis presented in Theorem 2. According to the mathematical framework, the critical transition occurs when the memory decay and context window effects achieve balance, formally expressed through the equilibrium condition:

$$\varepsilon(W, \lambda, t) = C^1/W + C^2\lambda^t \tag{7}$$

When λW approaches unity, the relative contributions of context limitations (C_1/W *term*) and memory decay ($C_2\lambda^t$ *term*) reach equilibrium, creating the phase transition phenomenon that distinguishes context-constrained coordination from classical game-theoretic scenarios. Statistical validation confirms that this mathematical prediction accurately captures the empirical behavior of real coordination systems.

The close alignment between theoretical predictions and statistical analysis demonstrates the descriptive accuracy of the CCNE framework while providing objective criteria for regime classification in practical system deployment. This correspondence between mathematical theory and empirical observation establishes confidence in the framework's ability to predict system behavior across different operational conditions and provides quantitative guidelines for system configuration based on specific performance requirements and resource constraints.

To validate the three-regime classification predicted by theory, partitioned the experimental data into groups based on theoretically motivated λW thresholds and conducted analysis of variance to test for significant performance differences. The grouping criteria were established as follows: Memory-Limited regime ($\lambda W < 0.9$, n = 83 experimental configurations), Context-Limited regime ($0.9 \leq \lambda W \leq 1.2$, n = 91 configurations), and Unconstrained regime ($\lambda W > 1.2$, n = 76 configurations).

The overall ANOVA revealed highly significant differences among the three operational regimes ($F(2,247) = 89.4$, $p < 0.001$, $\eta^2 = 0.42$), indicating that the regime classification explains 42% of the variance in coordination performance. Post-hoc analysis using Tukey's Honestly Significant Difference test confirmed that all pairwise comparisons between regimes were statistically significant ($p < 0.01$), validating the theoretical prediction that these represent qualitatively distinct operational modes.

The group means to provide quantitative characterization of each regime's performance characteristics. The Memory-Limited regime exhibited coordination success rates of 58.3% ± 3.2%, reflecting the substantial performance penalties imposed by insufficient memory capacity relative to context requirements. The Context-Limited regime achieved improved performance at 76.4% ± 2.8%, demonstrating the benefits of adequate memory capacity despite context window constraints. The Unconstrained regime reached 91.7% ± 1.9% success rates, approaching theoretical performance limits when neither memory decay nor context limitations impose binding constraints.

The empirically identified breakpoint at $\lambda W = 1.02$ provides remarkable correspondence with the theoretical threshold derived from the deviation bound analysis presented in Theorem 2. According to the mathematical framework, the critical transition occurs when the memory decay and context

TABLE 3. Operational regime analysis.

Regime	λW Range	W Range (tokens)	Performance Range (%)	Primary Limiting Factor
Memory-Limited	< 1.0	Any	42.3 - 58.7	Memory decay rate
Context-Limited	> 1.0	< 6000	58.7 - 84.2	Context window size
Unconstrained	> 1.0	> 6000	84.2 - 94.1	Task complexity

window effects achieve balance, formally expressed through the equilibrium condition in Equation (8):

$$\varepsilon(W, \lambda, t) = C_1/W + C_2\lambda^t \tag{8}$$

When λW approaches unity, the relative contributions of context limitations (C_1/W *term*) and memory decay ($C_2\lambda^t$ *term*) reach equilibrium, creating the phase transition phenomenon that distinguishes context-constrained coordination from classical game-theoretic scenarios. Statistical validation confirms that this mathematical prediction accurately captures the empirical behavior of real coordination systems.

The close alignment between theoretical predictions and statistical analysis demonstrates the descriptive accuracy of the CCNE framework while providing objective criteria for regime classification in practical system deployment. This correspondence between mathematical theory and empirical observation establishes confidence in the framework's ability to predict system behavior across different operational conditions and provides quantitative guidelines for system configuration based on specific performance requirements and resource constraints.

The statistical validation presented here transforms the qualitative regime boundaries suggested by visual inspection into rigorously validated, quantitative thresholds that can guide practical system design. The convergence of theoretical predictions, visual patterns, and statistical analysis provides compelling evidence for the fundamental nature of the phase transition phenomena identified in this work, establishing the $\lambda W \approx 1$ threshold as a reliable design parameter for context-constrained multi-agent coordination systems.

Table 3 quantifies the regime boundary analysis across different experimental conditions. The boundaries align closely with theoretical predictions, with the memory-context threshold occurring at $\lambda W = 1.08 \pm 0.15$ and the context saturation threshold occurring at $W = 5847 \pm 234$ tokens.

D. COMPARATIVE FRAMEWORK ANALYSIS

To establish the significance of the theoretical contributions, the analysis utilized Bonferroni-corrected t-tests with existing multi-agent coordination frameworks. Table 4 presents quantitative comparisons across multiple performance dimensions and system characteristics.

TABLE 4. Quantitative comparisons.

Framework	Success Rate (%)	Resource Efficiency	Scalability	Reproducibility	Overall Score
AutoGen	67.3 ± 3.8	0.54 ± 0.09	6.2	4.8	5.7
ChatDev	71.2 ± 3.5	0.61 ± 0.08	6.5	5.9	6.1
MetaGPT	74.8 ± 3.2	0.68 ± 0.07	7.1	6.8	6.5
CCNE Framework	89.7 ± 2.1	0.88 ± 0.04	8.2	9.1	8.6

Note: All metrics scaled so higher values indicate better performance.

The Context-Constrained Nash Equilibrium framework achieves superior performance across all evaluation dimensions. The 23% improvement in success rates over the best baseline framework represents a substantial advancement in coordination effectiveness. The 31% improvement in resource efficiency demonstrates that theoretical optimization principles translate directly into practical system improvements.

Metric Definitions: 1. Success Rate (%): Percentage of trials achieving task completion criteria per benchmark. Simple tasks: ≥90% subtask completion; Complex tasks: ≥75% subtask completion. Measured over 50 independent trials per configuration. 2. Resource Efficiency (0-1): Computed as (Success Rate) / (Token Usage × Time), normalized by baseline framework. Higher values indicate more efficient resource utilization. 3. Scalability (1-10): Measures performance degradation as agent count increases from 2 to 10. Score = 10 - (degradation_rate × 10), where degradation_rate is the slope of performance decline. 4. Reproducibility (1-10): Consistency across random seeds. Score = 10 × (1 - σ), where σ is the normalized standard deviation of success rates across 5 random seeds (0, 42, 123, 456, 789). 5. Overall Score (1-10): Weighted composite metric computed as: 0.4 × (Success Rate/100) × 10 + 0.25 × (Resource Efficiency) × 10 + 0.2 × Scalability + 0.15 × Reproducibility.

Table 4. Metrics (↑ higher is better). Success, Resource Efficiency (0–1), Scalability (1–10), Reproducibility (1–10), Overall (1–10).

The theoretical foundation score reflects the mathematical rigor and predictive capability of each framework. Existing frameworks rely primarily on heuristic approaches or predefined organizational structures, while the approach provides formal mathematical foundations with proven convergence guarantees and quantitative performance predictions.

VII. DISCUSSION AND IMPLICATIONS

The development of the Context-Constrained Nash Equilibrium (CCNE) and Linguistic Uncertainty Game (LUG) frameworks advances the understanding of strategic coordination by providing mathematical tools for modeling bounded-memory agents and analyzing semantic uncertainty in modern AI systems. These theoretical contributions establish new mathematical structures that capture

previously unmodeled aspects of multi-agent coordination under realistic constraints. Under practical conditions—such as asynchrony, limited memory and compute, and zero-trust boundaries—the frameworks are further supported by convergence and tracking guarantees from consensus control [23] and privacy-preserving distributed estimation in mobile robot networks [24], reinforcing their generality and extensibility.

The CCNE framework offers the first rigorous mathematical treatment of strategic interaction under bounded memory and computational constraints, addressing a key limitation in classical game theory, which assumes unlimited cognitive resources. The existence proofs, convergence analysis, and deviation bounds established here provide fundamental tools for designing bounded-rationality coordination systems. The LUG framework fills an equally critical gap at the intersection of game theory, natural language processing, and multi-agent systems. By formalizing strategic communication under semantic ambiguity, it bridges previously disconnected research areas and introduces new theoretical instruments for analyzing communication-based coordination. The robust communication equilibrium and information-theoretic communication bounds offer practical guidance for designing reliable coordination protocols.

As LLM capabilities continue to evolve, the analyzed constraints—context windows and semantic uncertainty—remain central to coordination challenges. The frameworks provide enduring principles applicable across diverse architectures, ensuring continued relevance. The identification of operational regimes offers concrete design guidelines for resource allocation and system configuration under specific deployment requirements. Systems operating below the critical threshold ($W^* \approx 4096$ tokens) face significant performance losses that cannot be compensated by other optimizations, implying that coordination architectures should target context windows of at least 6000–8000 tokens to operate effectively in the unconstrained regime. The strong negative correlation ($r = -0.85$) between semantic clarity and coordination success, validated across multiple experiments, offers predictive power for optimizing communication strategies that balance clarity and efficiency.

The non-monotonic learning dynamics observed in the analysis highlight that early performance degradation naturally occurs during advanced uncertainty modeling, guiding system designers to avoid prematurely discarding theoretically superior approaches. These insights translate directly into safety and robustness implications for multi-agent AI systems. Context-window constraints naturally bound the complexity of coordination strategies, enhancing predictability and safety despite limiting peak performance in some settings. The mathematical characterization of these bounds enables systematic failure-mode analysis prior to deployment. The robust communication equilibrium further mitigates coordination risks arising from semantic misinterpretation, while the operational regime analysis helps identify safe configuration regions across varying parameters.

The theoretical foundations developed here open several promising directions for future research in large-scale coordination, approximation algorithms for tractable CCNE computation, and extensions to multimodal communication involving text, vision, and audio. Incorporating human participants introduces additional bounded-rationality considerations, yet the same principles of constrained coordination remain applicable. As LLM architectures evolve toward dynamic context management and selective attention, the mathematical principles presented here offer a foundation for analyzing coordination under adaptive memory mechanisms. Overall, this work demonstrates how rigorous theoretical analysis, validated through production-level implementations, can directly inform system design and enable predictable, efficient, and safe coordination in increasingly complex multi-agent AI environments.

VIII. CONCLUSION

This research develops mathematical frameworks for analyzing strategic coordination under context-window constraints and semantic uncertainty in large-language-model-based multi-agent systems, addressing fundamental limitations that distinguish these systems from classical game-theoretic settings. Through the formulation of the Context-Constrained Nash Equilibrium (CCNE) and Linguistic Uncertainty Game (LUG) frameworks, the work extends mathematical analysis of strategic interaction to encompass bounded-memory limitations and semantic uncertainty that characterize realistic AI coordination scenarios.

The mathematical contributions include rigorous existence proofs, convergence analysis, and performance bounds that enable quantitative prediction of coordination behavior under practical constraints. Theoretical analysis reveals previously unknown phenomena—phase transitions in coordination effectiveness, systematic relationships between semantic uncertainty and performance, and operational-regime boundaries that offer concrete design guidance. These results transform qualitative observations about coordination challenges into quantitative relationships that enable principled optimization and reliable performance prediction.

Empirical validation demonstrates strong correspondence between theoretical predictions and observed system behavior, confirming that the proposed frameworks capture essential characteristics of real coordination systems. The predicted context-window phase transition at approximately 4,096 tokens, the substantial semantic-uncertainty effects observed in experiments, and the convergence-rate bounds all receive consistent experimental support across diverse coordination benchmarks.

The practical implications extend beyond academic contribution to actionable guidance for next-generation AI system design. Identification of critical context-window thresholds informs resource-allocation decisions, while quantitative relationships between semantic uncertainty and coordination effectiveness support principled communication-protocol design. Operational-regime analysis provides clear

guidelines for parameter selection based on deployment requirements and performance objectives.

The broader significance of this work lies in demonstrating that mathematical analysis can yield practical insights for advanced AI systems operating under realistic constraints. The successful integration of game theory, machine learning, and natural-language-processing methodologies illustrates the value of interdisciplinary approaches that combine theoretical rigor with empirical validation in production environments. As AI systems become increasingly complex, such integrated analysis will be essential for ensuring predictability and safety.

Future research directions include scaling the frameworks to large-scale coordination with hundreds or thousands of agents, developing approximation algorithms for tractable CCNE computation, and extending the theory to multi-modal and human-AI coordination scenarios. Incorporating dynamic context management and evolving agent capabilities represents another promising direction. The reproducibility guarantees—including machine-verifiable proofs and containerized experimental environments—ensure that these theoretical results can be independently validated and extended by the research community.

Overall, this work establishes rigorous mathematical foundations for understanding and designing coordination in LLM-driven multi-agent systems. By linking theoretical analysis with empirical validation, it provides both scientific insight and practical tools for developing effective, robust, and predictable coordination architectures in next-generation artificial-intelligence systems

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